Segment focused portfolio optimization based on machine learning predictions

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Motivation

- Noisy and high-frequency data generated on the stock exchange markets have been point of interest for decades.
- Being able to trust-worthy predict day-to-day changes of stock assets would be extremely beneficial for the industry – more informed decision in portfolio management, strategy evaluations, etc.
- There are quite a few studies involing both classification and regression machine learning (ML) approaches to predict developments on the market; however, many of them apply ML techniques only on very limited number of stocks [1, 2] or on stock indices [3].
- What is more is lack of studies applying prediction modelling in portfolio strategy optimization.

- This study evaluates prediction capabilities of selected ML algorithms and portfolio optimization framework on segment of **Information technology** (IT) among 500 biggest US companies by market cap during the period between September 2017 and January 2026.

Features for ML modelling

- The amount of variables, which are directly or indirectly generated with day-to-day market running is enormous and qualified selecting is necessary. Selection methodology in this study is based of *a priori* expectations of link between market and particular feature.

- Among **stock indices**, S&P 500 and NASDAQ Composite (NASDAQ) were chosen for their presumable prediction valuability.
- From the set of raw goods – **commodities** – traded on the market, gold and crude oil were selected.
- Lastly, data regarding **foreign exchange market** – exchange rates of selected major currencies (with American dollar), were considered.

- Apart from these, data quantifying **volatility** on the market and price developments of selected **US Treasury Bonds** were used.

Prediction results & Benchmarking

- We adopted **rolling window** framework for next-day closing price prediction of stocks. Data from trading days $t-q, t-(q-1), \ldots, t-1$ are employed as predictors for the closing price on day t.
- Several studies lack qualified benchmarking of obtained predictions to some naïve or low-effort solutions. In accordance with Fama [4], we employed benchmarking naïve prediction, assuming price on next day is equal to the yesterday's one.

Dominance corresponds to the fraction of predictions in average, when tuned model was more precise than naïve benchmark. Results of predictive hyper-tuned predictive models (on test sample) are as follows:

Model Type	MAPE (%)	Dominance (%)
Naïve benchmark	1.729	
ARIMA	1.923	56.68
Ridge (Stock-wise mode)	1.794	45.94
Ridge (General model)	1.798	45.88
Lasso (General model)	2.240	38.83
Lasso (Stock-wise model)	2.310	37.10

Out-performing naïve benchmark is extremely difficult; however, ARIMA model was able to out-perform it in dominance.

Model-backed portfolio optimization

- Obtained models are capable of reinforcing portfolio optimization frameworks (models are used to define expected returns). In the case of this study, modified **Markowitz mean-variance portfolio** was chosen [5].
- Employed modified version takes into account **transaction costs**.
  Let $\mathbf{w}_t, \mathbf{w}_{t-1}$ be assets weights in days $t, t-1$ respectively; $\boldsymbol{\mu}_t$ expected returns in day t; $\boldsymbol{\Sigma}$ covariance matrix and λ_R, λ_T hyperparameters of risk-aversion and transaction unit cost respectively; then capital allocation optimization is given as:

$$\max_{\mathbf{w}_t} \quad \mathbf{w}_t^\top \boldsymbol{\mu}_t - \lambda_R \mathbf{w}_t^\top \boldsymbol{\Sigma} \mathbf{w}_t - \lambda_T \|\mathbf{w}_t - \mathbf{w}_{t-1}\|_1$$
$$\text{Subject to:} \quad \mathbf{w}_t \geq \mathbf{0}_a, \quad \mathbf{1}_a^\top \mathbf{w}_t = 1.$$

Tuned portfolio models evaluations, on the interval between January 2023 and January 2026 (Testing interval), are as follows:

Portfolio Type	Annual Return (%)	Annual Volatility (%)
Naïve	16.48	25.20
ARIMA	15.67	25.48
Ridge	15.20	25.29
Lasso	16.88	25.53
S&P 500	8.35	23.10
NASDAQ	18.51	25.15

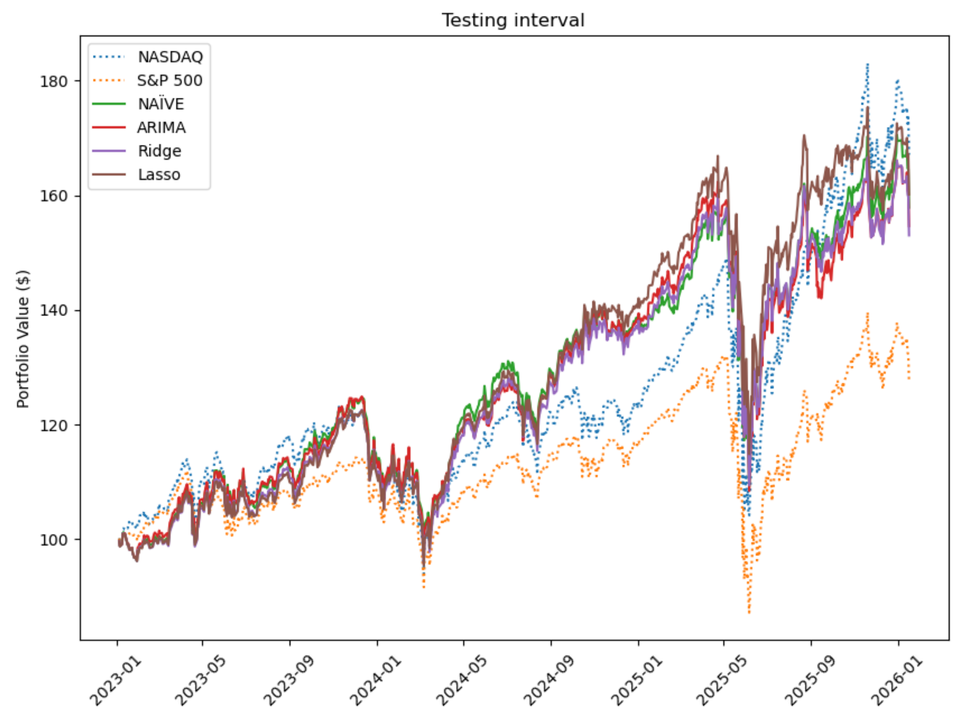

Figure 1: Performance of $100\,\$$ investment in proposed model and indices (Testing interval).

Conclusions & Discussion

- Time-dependence aware model (e.g. ARIMA) are capable of delivering comparatively better results, rather than conventional ML solutions.
- In order to build resilient investment strategy, extremely precise predictions of returns are not necessarily essential.

- During market phase of neither bull or bear (Training-validation interval), proposed approach involving prediction models is probably capable of delivering above-average results; however, during bull periods (2023 – 2026, AI surge), strategy could not compete with NASDAQ benchmark. Even though, general market (S&P 500) was beaten (in terms of annual return).
- Resulting data-pipeline is universal, capable of analyzing any set of stocks, segment, etc.

KEY REFERENCES

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Original research done for bachelor thesis. Further sources are saved as a Git repository.

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